



Optimization-based innovative approach for economical design of reinforced concrete isolated footings

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Abstract

Breaking away from conventional pad footing designs, stepped and sloped footings provide an innovative approach to isolated footing design. These alternative footings showcase a novel configuration that emphasizes eco-efficiency, surpassing traditional isolated pad footings in terms of sustainable and economical construction. Yet, optimization-based research on these designs remains nonexistent. This study aims to fill that gap by focusing on stepped footing optimization. A meta-heuristic genetic algorithm is used to optimize the design of isolated footings using this previously unexplored stepped footing approach, with the goal of maximizing cost savings. Two case studies from the literature are used to demonstrate the effectiveness of the adopted methodology. A comparative analysis is performed between the conventional designs and their respective optimized pad and stepped footing designs. The results show that the stepped footing optimization led to a significant reduction in concrete and reinforcement, resulting in an up to 16.24% more economical section than the optimized pad footing design. Apart from cost, two environmental factors, namely carbon emissions and embodied energy, have also been included as objective functions. It is revealed that the traditionally used carbon objective results in increased emissions and higher costs compared to the superior embodied energy objective.

Keywords Genetic algorithm · Isolated footings · Stepped footings · Optimized design · Economical design · Eco-efficient design

Introduction

The shift towards cost optimization and waste reduction in construction has been driven by increasing demands to mitigate climate change. Global greenhouse gas emissions have risen for the third consecutive year, reaching a record high of 52.4 gigatons of CO₂ equivalent [1]. The building industry is responsible for 30% of all greenhouse gas emissions, 40% of global resource use, and 40% of energy consumption [2, 3]. Concrete production has surged to 10 billion m³ [4],

contributing to increased costs, material waste, and carbon footprints in construction projects. Therefore, optimizing reinforced concrete (RC) structures is critical. This need has led to a new avenue of research, structural design optimization, which enables engineers to develop objective-based solutions for complex real-world challenges [5].

Optimization can target various objectives, but the primary focus remains on cost savings through material optimization [6–9], as reducing material costs is a key driver in RC optimization [8]. Consequently, linear [10, 11], non-linear [12–15], and meta-heuristic methods [16–21] have been widely employed, with the latter being particularly suitable for civil engineering structures [22–24]. Despite their effectiveness in economizing various RC elements [25–27], research on the optimization of RC footings is relatively scarce [8, 23, 24]. Existing studies have focused primarily on isolated pad footings [13–21, 28, 29], leaving alternative designs underexplored.

Isolated footings are the most commonly used foundation type for small to medium-sized projects. These shallow spread footings distribute column loads onto the soil and

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must resist structural, geotechnical, and equilibrium failures. Conventional isolated footings, or "isolated pad footings," often lead to unnecessary increases in size, costs, material waste, and carbon footprints. In these designs, critical sections near the column face are taken as demand loads for the entire footing. Although commonly used in practice, isolated pad footings fail to account for the decreasing demand-to-capacity ratio toward the edges, resulting in oversized designs, inflated costs, and increased material waste. To address these issues, innovative alternatives like stepped and sloped footings have been proposed. Stepped footings, in particular, introduce "steps" in areas of reduced load intensity, allowing for a reduction in footing depth and significant cost and volume savings. However, despite their potential, research on the optimization of isolated stepped footings remains lacking [30].

Current studies have predominantly focused on conventional pad footings, with little attention given to alternatives like stepped and sloped footings, which offer considerable potential for material and cost savings. Furthermore, their optimization for eco-efficiency—particularly in reducing emissions and embodied energy—remains unexplored. This gap underscores the need for further research into innovative footing designs that can both reduce costs and improve environmental sustainability.

This study aims to address these limitations by introducing an optimization approach for isolated stepped footings using a genetic algorithm (GA). By optimizing the geometric and reinforcement variables of stepped footings, the research seeks to improve both economic efficiency and eco-efficiency. Unlike previous studies that applied constant sections across pad footings, the stepped design offers more flexibility, enabling significant material and cost savings. The authors previously optimized the cost of isolated pad footings [18] as part of ongoing research on the economical design of structures at the University of Engineering and Technology, Lahore [31–33]. Building on this work, the current study introduces two environmentally conscious objective functions—carbon emissions and embodied energy—marking a notable contribution to eco-efficient structural design. The latter objective function is a unique application of this study and could prove more beneficial for the eco-efficient design of footings [34–36].

Building on the preceding discussion, the study is divided into two phases. In the first phase, isolated stepped footings are optimized for the most economical design using the genetic algorithm (GA). A previously developed spreadsheet tool by Waheed et al. [18] has been used, with modifications to incorporate variables and constraints specific to stepped footings. The design is conducted in compliance with ACI 318-14 [37] on two case study isolated footings. The results are compared with both the conventional design and the optimized pad footing design obtained by Waheed et al. [18]. In

the second phase, the tool is further modified to optimize the stepped footings for eco-efficiency, comparing three single-objective functions—cost, carbon emissions, and embodied energy. The novelty of this study lies in introducing stepped footings into the optimization process to maximize both cost and eco-efficiency. Additionally, this research expands the limited knowledge on using embodied energy as an objective function for structural design optimization, thereby contributing to eco-efficient practices in the field.

Methodology

The research methodology in this study builds on a previously developed spreadsheet-based tool for the design optimization of isolated pad footings using the evolutionary algorithm (EA) [18]. For this study, two key modifications have been introduced. First, the EA has been replaced by a more reliable and robust genetic algorithm (GA) to optimize the geometric and reinforcement parameters of the isolated footings. Second, the design problem has been adjusted to incorporate the unique approach of stepped isolated footings, requiring significant changes to the original tool. To maintain compatibility with the user-friendly spreadsheet interface, the MS Excel-compatible Evolver add-in tool was used to implement the GA [38]. The formulation of the optimization problem, including the objective function, variables, constraints, and settings for the GA, is discussed in Sect. "[Formulation of optimization problem for economical design](#)". This section also presents two case studies with both concentric and eccentric loading conditions, providing the results, discussion, and a comparative analysis between conventional footing designs, optimized pad footings, and optimized stepped footings. Finally, Sect. "[Formulation of optimization problem for eco-efficient design](#)" addresses the optimization of isolated stepped footings for eco-efficient design.

Design of single-stepped isolated footings

The design is mainly governed by the net allowable soil pressure, axial loads, and moments. Footings must be designed to resist geotechnical, structural, and equilibrium failures. Bending, shear, and bond failure are all examples of structural failures. Sliding, settlement, and bearing failure are included in geotechnical failures, while eccentricity problems resulting in overturning are included in the equilibrium failure. The design in this study is done in accordance with ACI 318-14 [37]. The following section details the design process of the stepped isolated footing as included in the problem formulation.

Service and factored soil pressure

To ensure stability, the maximum allowable contact pressure under service loads (q_{max}) should not exceed the net allowable bearing pressure of soil (q_{net}). The footing area ($L \times B$) is calculated considering service dead loads and service live loads (P_D, P_L), the service load moments (M_B, M_L) and the net allowed bearing pressure of soil (q_{net}). On the other hand, factored loads (P_u), and factored moments (M_{uB}, M_{uL}) are used for the structural design. To determine the shear and bending moments along the critical sections (Fig. 1) of the single-stepped footing, factored soil contact pressures are used. To avoid any tension or uplifting under the footing base, eccentricity along length (e_L) and width (e_B) is kept within one sixth of the respective dimensions of the footing.

$$\left\{ \begin{array}{l} \frac{P}{A} \pm \frac{M_B \left(\frac{B}{2} \right)}{I_B} \pm \frac{M_L \left(\frac{L}{2} \right)}{I_L} \\ q_{net} q_{max} \\ q_{min} 0 \\ e_L \leq L/6 \\ e_B \leq B/6 \end{array} \right. \quad (1-5)$$

Depth of footing

Shear and flexure criteria are typically used to calculate the depth of footing at the column face (d_1) and at the step face (d_2) by taking into account critical sections at these respective locations.

Design for shear

The stepped isolated footing modeled in this study incorporates both one-way (V_u) and two-way shear (Vu') in longer and shorter directions of footing as per ACI 318-14 code requirements. Additionally, eccentric punching shear (Vu'') is also included in the design of the stepped isolated footing in this study. For one-way shear, the critical section is positioned at a distance of (de_1) from the column face and (de_2) from the step's face in both directions. For punching and eccentric shear, the critical section is considered at distances of ($de_1/2$) from the column's edge and ($de_2/2$) from the step's face (Fig. 1). The pressure distribution under the footing is depicted in Fig. 2. This contact pressure is used to calculate one-way and punching shear at critical sections under the factored loading. Here, q_u is the ultimate factored soil pressure.

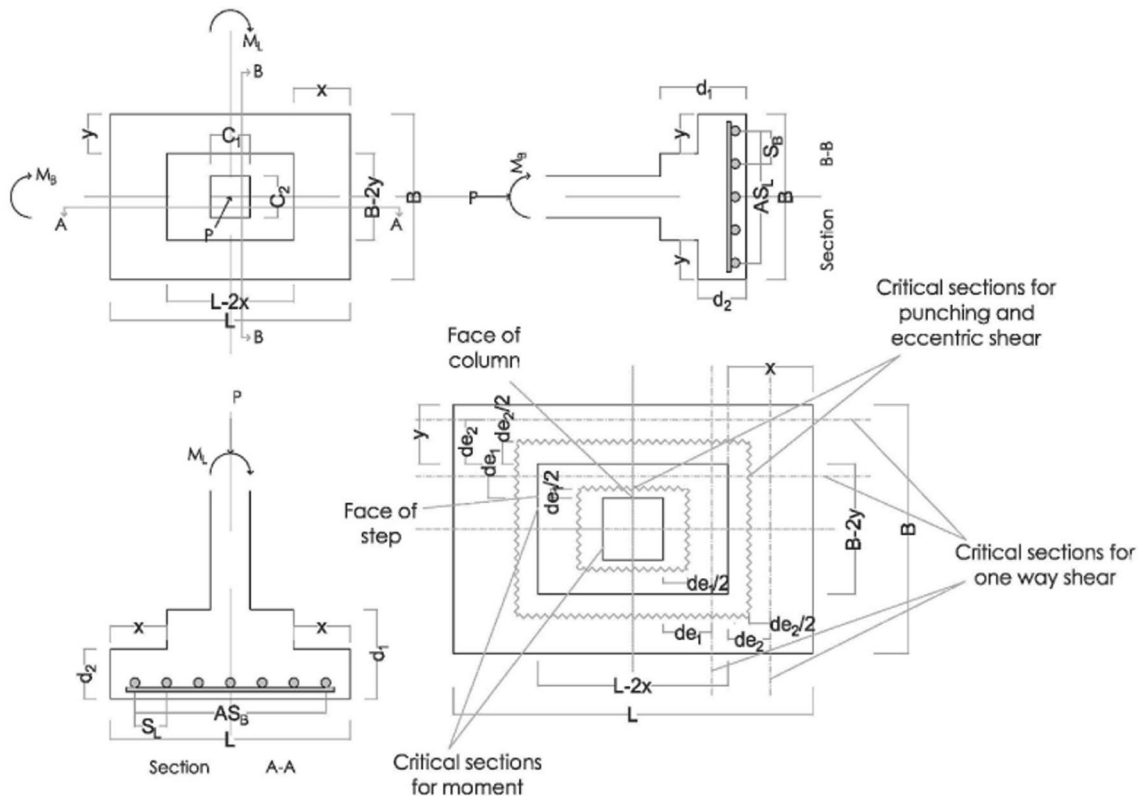


Fig. 1 Critical sections for moments and one-way, two-way and eccentric shear in reinforced concrete stepped isolated footing

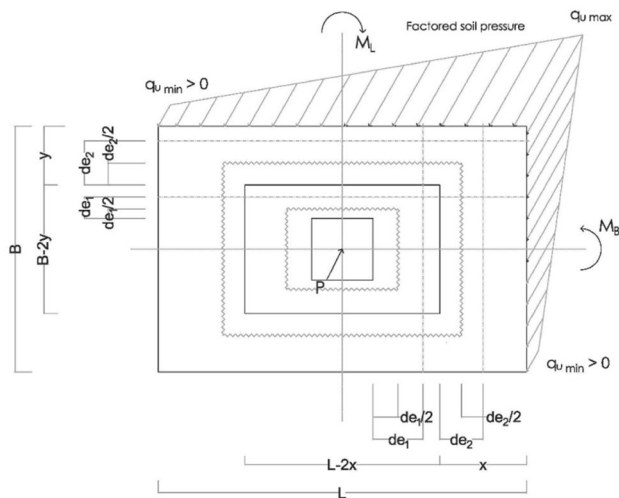


Fig. 2 Diagram illustrating pressure distribution in an eccentrically loaded footing under bi-axial bending

$$\left\{ \begin{array}{l} \phi V_n \geq V_u \\ V_n = \phi 0.17 \lambda \sqrt{f_c} b d_{e1or2} \\ V_n = \phi 0.33 \sqrt{f_c} \lambda b_o \frac{d_{e1or2}}{2} \\ V_n = \phi 0.17 \left(1 + \frac{2}{\beta} \right) \lambda \sqrt{f_c} b_o \frac{d_{e1or2}}{2} \\ V_n = \phi 0.083 \left(\frac{\alpha_s d_{e1or2}}{b_o} + 2 \right) \lambda \sqrt{f_c} b_o \frac{d_{e1or2}}{2} \end{array} \right. \quad (6-10)$$

Here, ϕ is the nominal strength coefficient, d_{e1} is the effective depth near the column's face and d_{e2} is the effective depth near the step's face, b is the unit length of a considered unit strip, β is the ratio of the longer to shorter side of the column, (α_s) is constant for column position, λ is the lightweight concrete modification factor (taken as 1 in this study), b_o is the critical section perimeter, f_y and f_c are steel yield strength and concrete compressive strength, respectively.

Design for flexure

The design of the base slab of footing is based on flexural analysis, where the bending moment at the critical section along both directions is taken into account (Fig. 1). The fundamental principle is to ensure that the capacity of the section (ϕM_n and ϕV_n) exceeds the demand at that section (M_u and V_u).

$$\left\{ \begin{array}{l} \phi M_n \geq M_u \\ \phi M_n = \phi A_s f_y \left(d_{e1or2} - \frac{a_{1or2}}{2} \right) \end{array} \right. \quad (11-12)$$

Here, ϕ is the nominal strength coefficient, A_s is the area of steel provided, d_{e1} is the effective depth near the face of the column, and d_{e2} is the effective depth near the face of the

step. Similarly, a_1 , a_2 are the depth of the compressive stress block near the face of the column and the face of the step, respectively.

Spacing of reinforcement and development length criteria

To ensure adequate resistance to slippage and bending moments, the ACI 318-14 [37] specifies guidelines for the maximum spacing (S_{max}), minimum clear spacing of reinforcement (S_{min}), and development length (l_{db}), which are crucial in creating the necessary tensile forces.

Formulation of optimization problem for economical design

The design optimization problem must be defined by formulating it in a mathematical form in an environment where an algorithm can be applied. The algorithm has eight user-defined variables and 25 constraints, and it is designed in accordance with the ACI 318-14 [37] building code. The units are kept in System International (SI).

Objective function

The total cost of the isolated footing is calculated based on V_c (volume of concrete), A_f (area of formwork), and W_{st} (weight of steel). The function $f(cost)$ for this study is defined in Eq. (13). Where, C_c is the cost of concrete per cubic meter, C_s is the cost of steel per kilogram, and C_f is the cost of formwork per square meter.

$$f(cost) = C_s W_{st} + C_c V_c + C_f A_f \quad (13)$$

Variables

The formulation takes into account eight variables that require both upper and lower limits. The limits on steel areas (A_{smin} and A_{smax}) and the minimum and maximum allowed footing thicknesses (d_1 , d_2) are obtained from ACI 318-14 [37] regulations. On the other hand, engineering judgement is used to determine the boundaries for the geometric variables (L , B , x , y). The variables are also depicted in Fig. 1, and Table 1 lists their lower and upper boundary limits. To demonstrate the effectiveness of the stepped footing over conventional isolated pad footing, the material strength characteristics are kept constant as they were presented by Waheed et al. [18].

Constraints

To ensure that the structure maintains proper functionality, limitations are imposed on the design parameters, restricting them within the allowable range specified by the building

Table 1 Design variables of stepped isolated footings and their limits

Variable	Units	Description	Lower Limits	Upper Limits
L	m	Length of the footing	1	10
B	m	Width of the footing	1	10
x	m	Length up to the step	0	10
y	m	Width up to the step	0	10
d_1	mm	Depth of footing at column face	150	1000
d_2	mm	Depth of footing at step face	150	500
As_L	mm ² /m	Area of steel along footing's length	As_{min}	As_{max}
As_B	mm ² /m	Area of steel along footing's width	As_{min}	As_{max}

code. This analysis takes into account a total of 25 restrictions, $g_1(x)$ - $g_{25}(x)$, relating to exterior stability, geometric feasibility, internal capacity, development lengths, eccentricity, and allowable reinforcement. These constraints are presented in detail in Table 2

Case studies

Two design examples, as also presented in Waheed et al. [18], have been used as case studies: the first by Wight and MacGregor [39] and the second by Kamara and Novak [40]. The analysis considers both concentric and eccentric loading conditions. Table 3 presents the input parameters and material unit rates used in the case studies. Specifically, Example 1 involves eccentric loading with moments, while Example 2 uses concentric loading. To evaluate the effectiveness of the stepped footing in comparison to a simple pad footing, the results are optimized and compared.

Results and discussion

As discussed above, the modified spreadsheet-based tool is utilized for the application of the optimization algorithm for two case study isolated footings. The algorithm is implemented by using the Evolver add-in tool [38]. The spreadsheet environment was used to input the design variables and parameters, and then the GA was launched separately from the Evolver's tab. The GA population size was set to 50, the mutation rate to 0.075, and the cross-over rate to 0.5, with a maximum change rate of 0.01% over 20,000 trials. The "recipe approach" was used for adjusting variables independently within predetermined constraints. The GA algorithm was used to obtain optimized results for the geometric (L , B , d_1 , d_2 , x , y) and reinforcement parameters (As_L and As_B). The optimized stepped footing results for both examples are shown in Tables 4 and 5, respectively. In addition to the stepped footing results, the tables also present the conventional and optimized pad footing designs that were developed using the same input parameters as Examples 1 and 2. The total cost and the percentage of cost savings resulting from all three design approaches are calculated at the end.

Example 1 was optimized through two runs, taking 190 s for the first and 129 s for the second run. The first run produced 47,388 trials, while the second generated 37,094. The optimized results showed a decrease in concrete volume from 7.45 to 3.56 m³ and steel usage from 208.48 to 167.27 kg. This led to a 41.77% reduction in overall cost. Comparatively, the optimized pad footing resulted in a reduction of 30.48% in cost. This means that an additional 16.24% of cost reduction was achieved using the innovative stepped footing approach.

Similarly, Example 2 was optimized through two runs, taking 103 s for the first and 114 s for the second run. The first run produced 44,024 trials, while the second generated 34,183. The optimized results showed a decrease in concrete volume from 10.18 to 6.64 m³ and steel usage from 364.95 to 294.20 kg. This led to a 28.78% reduction in overall cost. Comparatively, the optimized pad footing resulted in a reduction of 18.08% in cost. This means that an additional 13.06% of cost reduction was achieved using the stepped footing approach.

Figure 3 graphically represents the trend of gradual cost reduction obtained by the use of stepped footings by comparing the conventional design cost, optimized pad footing cost, and the stepped footing cost for both case studies. It can be inferred that the optimization aimed to lower the material quantity in both examples. Since steel is a more expensive material, reducing its quantity was more essential to cost reduction.

Further, trends are drawn in Figs. 4, 5 and 6 to observe the effects of optimization on volume of concrete, area of footing, and volume of steel. The volume of concrete decreased by 52.22% and 34.79%, the area of the footing decreased by 19.79% and 8.93%, and the weight of the steel decreased by 19.77% and 19.39%, for Examples 1 and 2, respectively.

Formulation of optimization problem for eco-efficient design

Objective function

As a novelty of this study, three objectives have been taken to be minimized, including cost, carbon emissions, and

Table 2 Design constraints of the stepped isolated footing

Constraint	Formula	Description
$g_1(x)$	$Vu_l + Vu_e \leq \phi V_c$	Concrete shear strength (ϕV_c) greater than maximum punching (Vu_l) and eccentric shear (Vu'') at distance $de_{1/2}$ from face of column
$g_2(x)$	$Vu_l + Vu_e \leq \phi V_c$	Concrete shear strength (ϕV_c) greater than maximum punching (Vu_l) and eccentric shear (Vu'') at distance $de_{1/2}$ from face of the step
$g_3(x)$	$Vu \leq \phi V_c$	Concrete shear strength (ϕV_c) greater than maximum one-way shear (Vu) at distance de_1 from the face of column along the length of footing
$g_4(x)$	$Vu \leq \phi V_c$	Concrete shear strength (ϕV_c) greater than maximum one-way shear (Vu) at distance de_1 from the face of column along the width of footing
$g_5(x)$	$Vu \leq \phi V_c$	Concrete shear strength (ϕV_c) greater than maximum one-way shear (Vu) at distance de_2 from the face of step along the length of footing
$g_6(x)$	$Vu \leq \phi V_c$	Concrete shear strength (ϕV_c) greater than maximum one-way shear (Vu) at distance de_2 from the face of step along the width of footing
$g_7(x)$	$M_{uB} \leq \phi_b Mn$	Flexural strength ($\phi_b Mn$) greater than maximum flexure moment about the axis perp. to footings width (M_{uB}) at the face of column
$g_8(x)$	$M_{uL} \leq \phi_b Mn$	Flexural strength ($\phi_b Mn$) greater than maximum flexure moment about the axis perp. to footings length (M_{uL}) at the face of column
$g_9(x)$	$M_{uB} \leq \phi_b Mn$	Flexural strength ($\phi_b Mn$) greater than maximum flexure moment about the axis perp. to footings width (M_{uB}) at the face of step
$g_{10}(x)$	$M_{uL} \leq \phi_b Mn$	Flexural strength ($\phi_b Mn$) greater than maximum flexure about the axis perp. to footings length (M_{uL}) at the face of step
$g_{11}(x)$	$de_1 < (\frac{L}{2} - \frac{c_1}{2} - x)$	Effective depth de_1 within the locality defined between the face of column and boundary of step along length of the footing
$g_{12}(x)$	$de_1 < (\frac{B}{2} - \frac{c_2}{2} - y)$	Effective depth de_1 within the locality defined between the face of column and boundary of step along width of the footing
$g_{13}(x)$	$de_2 < x$	Effective depth de_2 within the locality defined between the face of step and boundary of the footing along length of the footing
$g_{14}(x)$	$de_2 < y$	Effective depth de_2 within the locality defined between the face of step and boundary of the footing along width of the footing
$g_{16}(x)$	$q_{max} \leq q_{net}$	Net soil pressure (q_{net}) greater than maximum allowable contact pressure under service load incorporating moments (q_{max})
$g_{17}(x)$	$A_{req.} \leq A_{pro.}$	Max. footing area provided ($A_{pro.}$) greater than required footing area ($A_{req.}$)
$g_{18}(x)$	$e_L \leq L/6$	Eccentricity along length (e_L) less than one sixth of the length of footing
$g_{19}(x)$	$e_B \leq B/6$	Eccentricity along width (e_B) less than one sixth of the width of footing
$g_{20}(x)$	$As_{min} \leq As_B \leq As_{max}$	Max. (As_{max}) & min. (As_{min}) reinforcement along width
$g_{21}(x)$	$As_{min} \leq As_L \leq As_{max}$	Max. (As_{max}) & min. (As_{min}) reinforcement along length
$g_{22}(x)$	$l_{db} \leq \frac{L}{2} - Cov_f - d_b$	Development length along length (l_{db})
$g_{23}(x)$	$l_{db} \leq \frac{B}{2} - Cov_f - d_b$	Development length along width (l_{db})
$g_{24}(x)$	$S_{min} \leq S_L \leq S_{max}$	Spacing of steel along length (S_L)
$g_{25}(x)$	$S_{min} \leq S_B \leq S_{max}$	Spacing of steel along width (S_B)

embodied energy to maximize the eco-efficiency of isolated stepped footings. A truncated version of the previously taken cost-based objective function $f(cost)$ is used, as defined in Eq. (14).

$$f_{cost} = C_s W_{st} + C_c V_c \quad (14)$$

The second objective is minimum carbon emissions; it considers 4 factors, namely: e_c (carbon emission of concrete), V_c (volume of concrete), e_s (carbon emission of steel), and W_{st} (weight of steel). Equation (15) defines the carbon emissions-based function, $f(emissions)$.

$$f_{emissions} = e_s W_{st} + e_c V_c \quad (15)$$

The third objective is the minimum embodied energy; it also takes into account 4 factors, namely: E_c (embodied energy of concrete), V_c (volume of concrete), E_s (embodied energy of steel), and W_{st} (weight of steel). Equation 16 defines the embodied energy-based function, $f(energy)$.

$$f_{energy} = E_s W_{st} + E_c V_c \quad (16)$$

Values of both carbon emissions and embodied energy for concrete are extracted from the inventory of carbon

Table 3 Summary of input parameters for examples 1 and 2

Input Parameters	Example 1	Example 2	Unit
Concrete cover (Cov_f)	63.5	63.5	mm
Net allowable bearing pressure of soil (q_{net})	191.5	268	kPa
Dead load (P_D)	800.68	2406.49	kN
Live load (P_L)	533.79	862.95	kN
Dead load moment about the axis perpendicular to width (M_{DB})	0	0	kNm
Live load moment about the axis perpendicular to width (M_{LB})	0	0	kNm
Dead load moment about the axis perpendicular to length (M_{DL})	108.48	0	kNm
Live load moment about the axis perpendicular to length (M_{LL})	81.36	0	kNm
Column length (c_1)	406	610	mm
Column width (c_2)	406	610	mm
Reinforcement bar diameter along length (d_b)	22	25	mm
Reinforcement bar diameter along width (d_b)	22	25	mm
Compressive strength of concrete	24	28	MPa
Yield strength of steel	415	415	MPa
Unit rate of concrete	7460	7856	Rs/m ³
Unit rate of steel	120	120	Rs/kg

Table 4 Results of design optimization for example 1

Parameters	Conventional Pad Footing Design	GA Optimized Design of Pad Footing	GA Optimized Design of Stepped Isolated Footing	Units
Footings length (L)	3.66	3.1	2.86	m
Footings width (B)	3.05	2.86	3.17	m
Length up to first step (x)	—	—	0.685	m
Width up to first step (y)	—	—	0.832	m
Depth of footing at the face of column (d_1)	660.00	543.68	675.49	mm
Depth of footing after the step (d_2)	660.00	543.68	300.34	mm
Concrete compressive strength (f_c)	24	24	24	MPa
Steel yield strength (f_y)	415	415	415	MPa
Area of steel (As_L) along the footings length	1188	1307.28	1232.33	mm ² /m
Area of steel (As_B) along the footings width	1268	1246.69	1273.73	mm ² /m
Total optimized cost	84,590	58,807	49,259	PKR
% Optimization		30.48	41.77	%

and energy (ICE) database V2.0 [41] with respect to their strength and are detailed in Table 6. However, the values of steel are based on its type only, which in this case is reinforcing bar (virgin). Unit emissions of a reinforcing bar are taken as $1.99 \text{ kgCO}_2/\text{kg}$, and its embodied energy is taken as 35.4 MJ/kg .

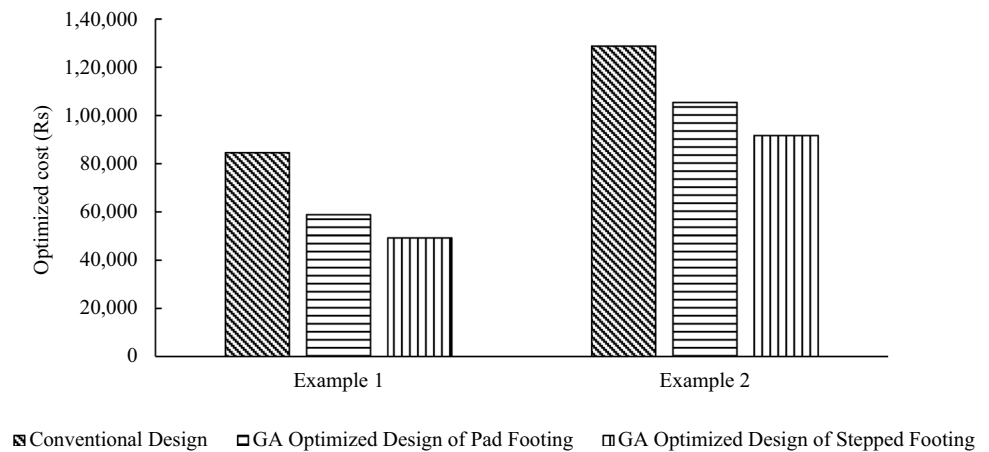
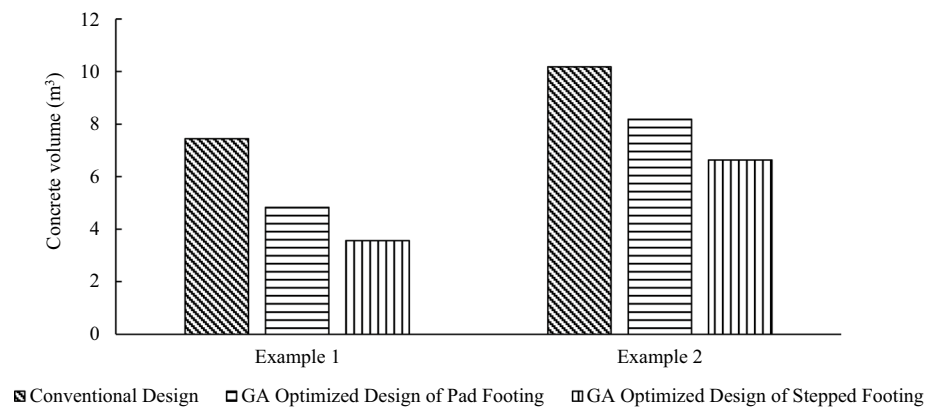
Implementation of eco-efficient design optimization

The goal of this section of the study is to assess three different single-objective functions for the RC stepped footing design that adheres to structural requirements while being the most environmentally friendly and economical. The case study isolated footings are again optimized as a

stepped footing to demonstrate the efficacy of the adopted methodology to maximize eco-efficiency. All units are in the SI system, and the initial values are taken from the previous Examples 1 and 2. The modified stepped footing tool is further adjusted to allow for cost, carbon emissions, and embodied energy optimization with a mutation rate of 0.1. Optimization is conducted for three different goals, namely, minimum cost, minimum carbon emissions, and minimum embodied energy. After optimization, the value of cost, emission, and energy is documented for each aim. For a better understanding of the optimization process, the value of steel and concrete contribution for each aim is also determined. In order to establish the most eco-efficient objective for a stepped isolated footing, GA

Table 5 Results of design optimization for example 2

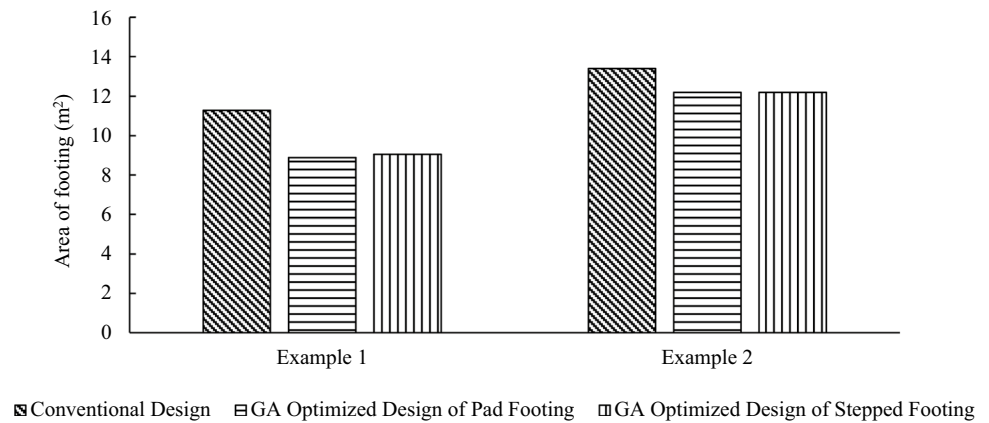
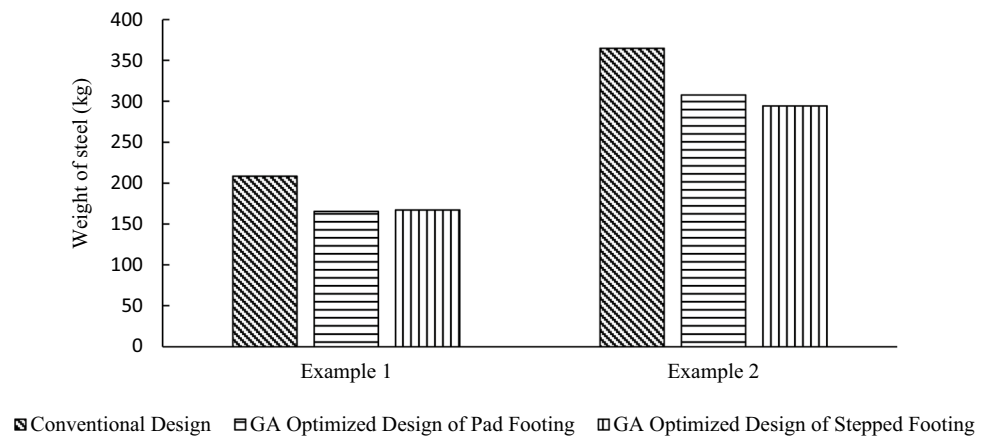
Parameters	Conventional Pad Footing Design	GA Optimized Design of Pad Footing	GA Optimized Design of Stepped Isolated Footing	Units
Footings length (L)	3.66	3.53	3.53	m
Footings width (B)	3.66	3.46	3.46	m
Length up to first step (x)	–	–	0.584	m
Width up to first step (y)	–	–	0.636	m
Depth of footing at the face of column (d_1)	760.00	671.44	881.66	mm
Depth of footing after the step (d_2)	760.00	671.44	296.97	mm
Concrete compressive strength (f_c')	28	28	28	MPa
Steel yield strength (f_y)	415	415	415	MPa
Area of steel (A_{sL}) along the footings length	1811.47	1800	1647.13	mm ² /m
Area of steel (A_{sB}) along the footings width	1811.47	1682.82	1608.87	mm ² /m
Total optimized cost	128,780	105,497	91,714	PKR
% Optimization		18.08	28.78	%

Fig. 3 Comparison of obtained cost for example 1–2**Fig. 4** Comparison of concrete volume required for example 1–2

optimization is performed for a total of three different scenarios on the above two examples. Additionally, the findings are represented graphically to show the level of optimization and the effects of optimization on the relevant material component.

Results and discussion

Same design Examples 1 and 2 are optimized again in the modified stepped footing spreadsheet, but this time for three different objectives. Comparisons are made with the conventional and optimized counterparts for evaluation of functions

Fig. 5 Comparison of footing area required for example 1–2**Fig. 6** Comparison of steel reinforcement required for example 1–2**Table 6** Unit carbon emissions of concrete with respect to concrete strength

Concrete compressive strength (MPa)	Carbon emissions (kgCO ₂ /m ³)	Embodied energy (MJ/m ³)
20	240	1680
25	256.8	1776
30	271.2	1872
35	288	1968
40	316.8	2112
50	362.4	2400

and to measure the effectiveness of the innovative stepped footing approach. Table 7 details the initial values of the geometric dimensions and steel reinforcement and the optimized values obtained by each run, for example 1. However, the degree of optimization will vary for each case as it is highly dependent on the original design parameters. It can also be further improved by relaxing the limiting bounds on the variables.

To better illustrate the above results, Fig. 7 is presented. It validates the algorithm's attempts in optimizing design

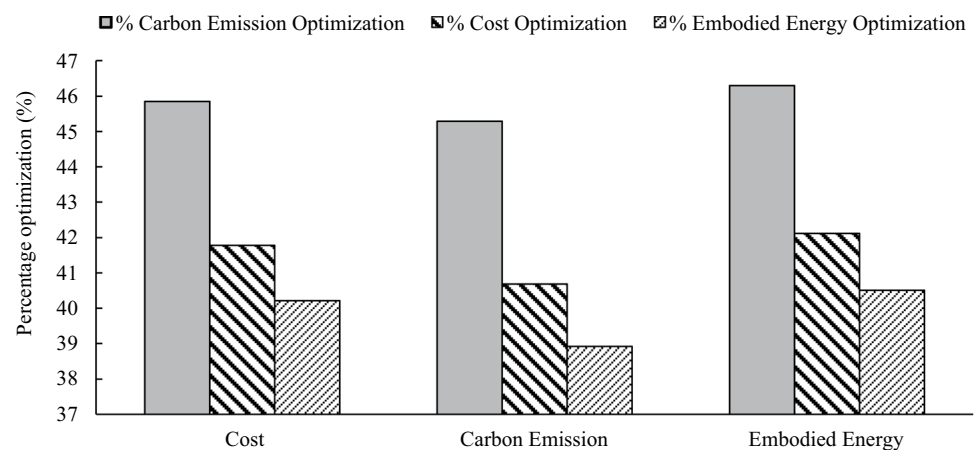
variables for the selected objective. Among the three cases, the total embodied energy objective has one of the best performances. Maximum optimization of 42.1% for total cost, 46.3% for carbon emissions, and 40.5% for embodied energy is achieved with this function. Least optimization is achieved by the minimum emissions objective, with the optimization of 40.7% for total cost, 45.3% for carbon emissions, and 38.9% for embodied energy. Table 8 details the initial values of the geometric dimensions and steel reinforcement and the optimized values obtained by each run, for example 2.

To better illustrate the above results, Fig. 8 is presented. Once again, the embodied energy objective has one of the best performances. With this objective, maximum optimization of 32.16% for total cost, 35.1% for carbon emissions, and 31.01% for total embodied energy is achieved. Least optimization is once again achieved using the minimum emissions objective, with optimization of 24.8% for total cost, 30.66% for carbon emissions, and 22.52% for total embodied energy.

From the illustration presented in Fig. 9, it can be concluded that cost and energy minimization have a synergy. This is because for cost and energy minimization, the optimization algorithm tries to reduce the quantity of steel as

Table 7 Optimized design variables for three objective functions for example 1

Parameters	Units	Conventional Pad Footing design	GA (objective functions)		
			Cost	Carbon Emission	Embodied Energy
Footing length (L)	m	3.70	2.93	2.98	2.89
Footing width (B)	m	3.05	3.07	3.01	3.13
Length up to first step (x)	m	–	0.70	0.72	0.69
Width up to first step (y)	m	–	0.69	0.73	0.81
Depth of footing at the face of column (d_1)	mm	660.00	692.37	703.42	682.89
Depth of footing after the step (d_2)	mm	–	285.53	293.53	298.59
Concrete compressive Strength (f_c')	MPa	24.00	24.00	24.00	24.00
Steel yield strength (f_y)	MPa	415.00	415.00	415.00	415.00
Area of steel (A_{sL}) along footings length	mm ² /m	1188.00	1247.01	1354.66	1246.63
Area of steel (A_{sB}) along footings width	mm ² /m	1268.00	1307.30	1332.93	1264.50

Fig. 7 Percentage optimization achieved for the three objective functions with their respective values of total cost, carbon emissions and embodied energy**Table 8** Optimized design variables for three objective functions for example 2

Parameters	Units	Conventional Pad Footing design	GA (objective functions)		
			Cost	Carbon Emission	Embodied Energy
Footings length (L)	m	3.66	3.21	3.02	3.28
Footings width (B)	m	3.66	3.80	4.04	3.72
Length up to first step (x)	m	–	0.64	0.50	0.66
Width up to first step (y)	m	0.00	0.67	0.53	0.72
Depth of footing at the face of column (d_1)	mm	760.00	801.44	752.90	819.64
Depth of footing after the step (d_2)	mm	–	309.83	266.08	321.54
Concrete compressive Strength (f_c')	MPa	28.00	28.00	28.00	28.00
Steel yield strength (f_y)	MPa	415.00	415.00	415.00	415.00
Area of steel (A_{sL}) along footings length	mm ² /m	1811.47	1505.92	1428.41	1523.40
Area of steel (A_{sB}) along footings width	mm ² /m	1811.47	1910.69	2327.23	1709.64

it is more expensive and has a much larger unit embodied energy than concrete. The embodied energy of steel is about 35–50 times the embodied energy of concrete, with a strength variation from 20 to 50 MPa. Contrarily, the unit carbon emission of steel is still higher than that of concrete, but only by about 13–20 times the unit carbon emission of

concrete for strength, which varied from 20 to 50 MPa. This places carbon optimization in a conflicting position with the other two objectives, leading to its comparatively poor performance. When compared to the most optimized result, the traditionally taken carbon objective provides 1.88% more emissions while increasing cost by up to 2.5%, for example

Fig. 8 Percentage optimization achieved for all objective functions with their respective values of total cost, carbon emissions and embodied energy

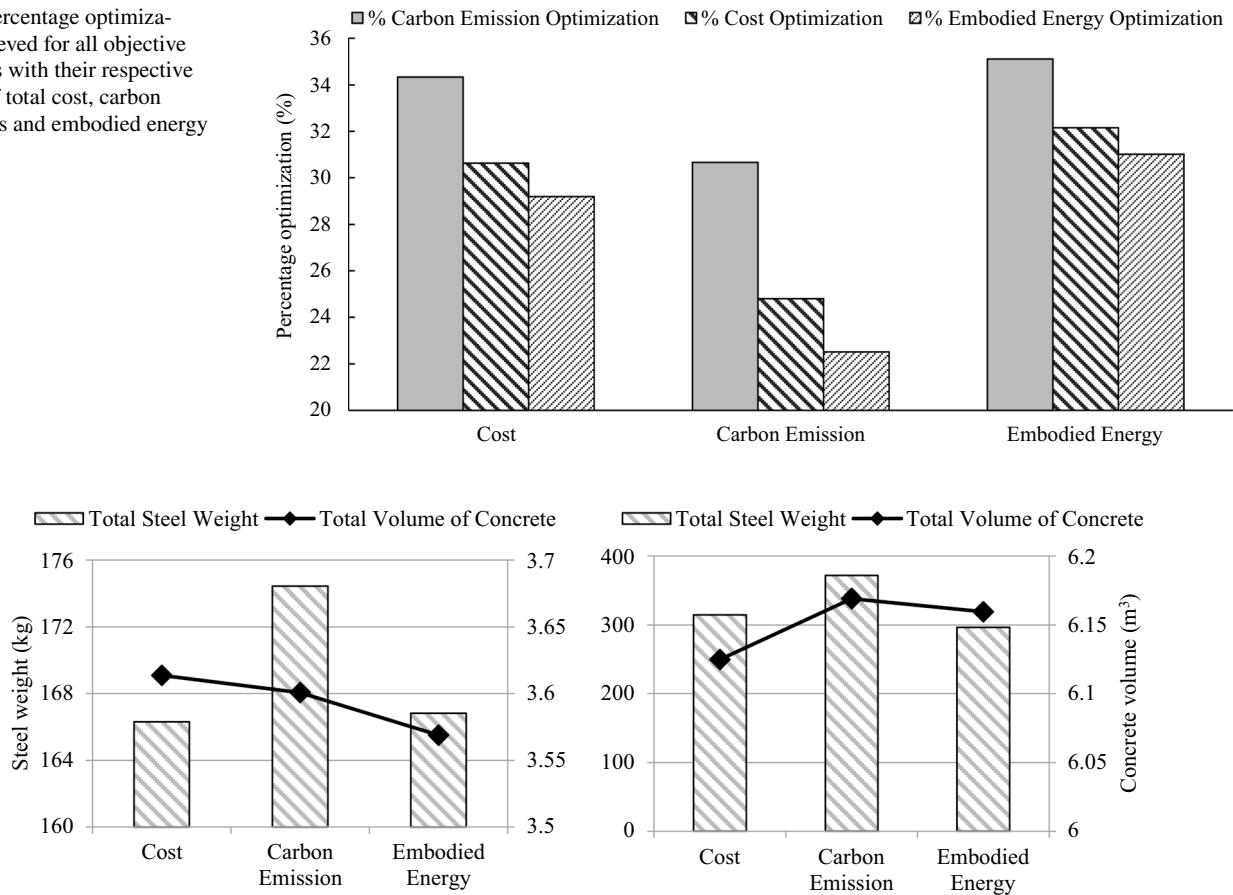


Fig. 9 Steel weight and concrete volume obtained with respect to each objective function for example 1 (left) and example 2 (right)

1. Similarly, it provides 6.84% more emissions than the other objectives while increasing cost by up to 10.84%, for example 2. This proves that the minimum carbon emissions function is not as effective for eco-efficiency as cost or embodied energy, for the undertaken case studies.

For the cost and embodied energy objectives, both steel weight and concrete volume are minimized compared to the carbon emission objective. The carbon emission-focused optimization leads to a higher steel requirement (174.4 kg) and concrete volume (3.6 m³) compared to cost and embodied energy minimization, for Example 1. Similarly, for example 2, carbon-focused optimization leads to a significantly higher steel requirement (371.78 kg) and concrete volume (6.17 m³) compared to other objectives. This demonstrates that while carbon emission reductions can be achieved, they come at the expense of increased material use, specifically in terms of steel, which is less efficient for both cost and embodied energy. Therefore, while carbon minimization objective has an environmental benefit, it results in a less optimized solution from an economic and material efficiency perspective. These results reinforce the trade-offs between these conflicting objectives in the optimization process.

Conclusions

In this study, a previously developed spreadsheet-based optimization tool for RC isolated footing design was modified to investigate the efficacy of a stepped isolated footing approach in maximizing economy. The results obtained were compared with conventional designs and optimized isolated pad footings using similar input parameters. The findings of the study are as follows:

- Optimizing pad footings using GA resulted in significant cost reductions of 30.48% and 18.08% for Examples 1 and 2, respectively.
- The incorporation of a single step in the optimized isolated footing design achieved greater cost savings, with reductions of 41.77% and 28.78% for Examples 1 and 2, respectively.
- The optimized isolated stepped footing was shown to be lighter than the conventional design, with reductions of 52.22% in concrete volume and 19.77% in steel weight

for Example 1, and reductions of 34.79% in concrete volume and 19.39% in steel weight for Example 2.

- Comparative study of cost, carbon emissions, and the novel embodied energy objective is performed for the most eco-efficient footing design.
- Results indicate that the commonly used emission objective, regresses the degree of optimization. In both Examples 1 and 2, the carbon objective yielded reductions in emissions but also increased total cost by 1.88% and 6.84%, respectively, compared to the most optimized design.

This study demonstrates that stepped footings, optimized through cost and embodied energy objectives, provide a significant advantage not only over conventional pad footing designs but also over optimized pad footings, paving the way for future innovations in eco-efficient footing design.

Recommendations for future works

While the current study demonstrates the potential of stepped footings for cost and material savings, several avenues remain for future exploration. Future studies could explore optimizing more complex configurations like multi-stepped or sloped footings, which may offer additional optimization and improved constructability. Expanding the analysis to different soil conditions, load cases, or footing geometries could further validate the robustness of the approach. Relaxing constraints by introducing soft constraints could increase design flexibility. Further investigating the effects of varying unit costs, carbon emissions, and embodied energy based on regional differences or material sources would provide more comprehensive eco-efficiency results. Finally, a comparative study between stepped and sloped footings would be valuable in order to determine the most optimized options for structural performance, cost-effectiveness, and eco-efficiency.

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Data availability The data used in this study has been presented and referenced within the article.

Declarations

Conflict of interest The authors have no conflicts of interests to declare that are relevant to the submitted work.

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